Question 1.1.1

Spring-Mass-Damper System: Rigorous Damping Analysis

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Problem Statement

Question:
A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007):

Options:

a) 2 Ns/m
b) 2 N/s
c) 0.2 kg/s
d) 0.2 N/m

System Model

Governing Differential Equation: By applying Newton's Second Law to an unforced spring-mass-damper system, we get the equation of motion:

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{1}$$

Where:

- m = mass (**1 kg**)
- c = actual damping coefficient (Unknown)
- k = spring stiffness

Derivation: Critical Damping (c_c)

Assuming a solution of the form $x(t) = Ce^{st}$, we obtain the characteristic equation:

$$ms^2 + cs + k = 0$$

Using the quadratic formula, the roots are:

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

Critical Damping Condition: Critical damping (c_c) occurs when the roots are real and equal, meaning the discriminant is zero:

$$c_c^2 - 4mk = 0$$

$$c_c = 2\sqrt{mk}$$

Derivation: Natural Frequency & Damping Ratio

The undamped natural frequency (ω_n) is defined as:

$$\omega_n = \sqrt{\frac{k}{m}} \implies k = m\omega_n^2$$

Substituting k back into the critical damping equation:

$$c_c = 2\sqrt{m(m\omega_n^2)}$$
$$c_c = 2m\omega_n$$

The **damping ratio** (ζ) is defined as the ratio of actual damping to critical damping. We are given $\zeta = 0.2$ and $\omega_n = 5$ rad/s:

$$\zeta = \frac{c}{c_c} = \frac{c}{2m\omega_n} \implies c = 2\zeta m\omega_n \tag{2}$$

Calculation & Dimensional Analysis

Step 1: Calculate the Magnitude

$$c = 2 \cdot (0.2) \cdot (1) \cdot (5)$$
$$c = 2$$

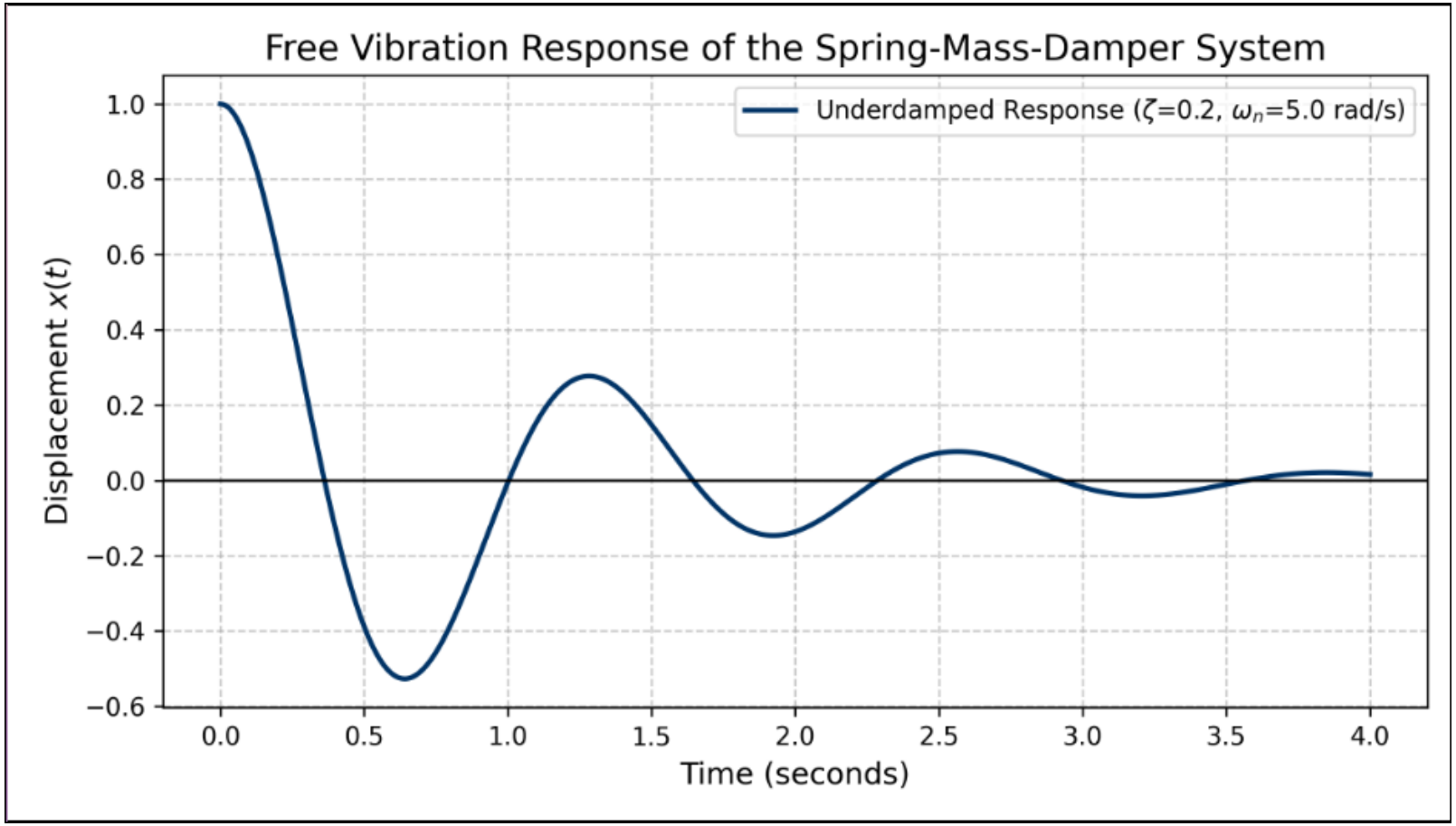
Step 2: Dimensional Analysis (Checking Units)
The damping force is proportional to velocity ($F_d = c \cdot \dot{x}$). Therefore, the units of c must be Force divided by Velocity:

$$[c] = \frac{\text{Force}}{\text{Velocity}} = \frac{\text{Newtons (N)}}{\text{Meters per second (m/s)}} = \mathbf{Ns/m}$$

Note: 1 N = 1 kg·m/s^2, so Ns/m is also equivalent to kg/s, but option (c) has the wrong magnitude.

System Response (Underdamped)

Since $\zeta = 0.2 < 1$, the system is **underdamped** and will oscillate before settling.



Conclusion

Based on the theoretical derivations and calculations:

- The magnitude is **2**.
- The correct engineering units for the damping coefficient are **Ns/m**.

Final Answer

The correct option is **(a) 2 Ns/m**.